

CAM – MIDTERM EXAM 06/11/2020

SOLUTIONS TO NUMERICAL PROBLEMS

TURNING

A steel bar with a diameter of 30 mm and a length of 160 mm is held by means of a self-centering three-jaw chuck and tailstock (dead centre). The workpiece undergoes external cylindrical turning, which brings the diameter to the final value of 24 mm over a length of 90 mm. The machining is performed in two steps: a roughing step and a finishing step. Considering the roughing step, whose depth of cut is equal to 2 mm, determine:

1. The values of cutting force and cutting power for the roughing step, taking into account a value of $k_{c0,4}$ equal to 2150 MPa, a coefficient x of the Kronenberg relationship equal to 0,29 and a value of main entering angle equal to 90° . The recommended feed is 0,15 mm/rev while the cutting speed is 260 m/min.
2. The optimal tool life and the corresponding cutting speed, in case you want to optimize the production time. The tool change time T_t is equal to 120 seconds, while the constants of the Taylor relation are $C = 250$ and $n = 0,122$.

Solution

Considering the roughing step, whose depth of cut is equal to 2 mm, determine:

1. The values of cutting force and cutting power for the roughing step, taking into account a value of $k_{c0,4}$ equal to 2150 MPa, a coefficient x of the Kronenberg relationship equal to 0,29 and a value of main entering angle equal to 90° . The recommended feed is 0,15 mm/rev while the cutting speed is 260 m/min.

$$k_c = k_{c0,4} \left(\frac{0,4}{f \cdot \sin k_r} \right)^x = 2150 \left(\frac{0,4}{0,15 \cdot \sin 90^\circ} \right)^{0,29} = 2.857,4 \text{ MPa}$$

$$A_D = f \cdot a_p = 0,15 \cdot 2 = 0,3 \text{ mm}^2$$

$$F_c = k_c A_D = 2.857,4 \cdot 0,3 = 857,2 \text{ N}$$

$$P_c = F_c v_c = 857,2 \cdot \frac{260}{60} = 3.714,6 \text{ W}$$

2. The optimal tool life and the corresponding cutting speed, in case you want to optimize the production time. The tool change time T_t is equal to 120 seconds, while the constants of the Taylor relation are $C = 250$ and $n = 0,122$.

$$T_{opt}(T_c) = T_t \left(\frac{1}{n} - 1 \right) = \frac{120}{60} \left(\frac{1}{0,122} - 1 \right) = 14,39 \text{ min}$$

$$v_{c,opt}(T_c) = \frac{C}{T_{opt}^n} = \frac{250}{14,39^{0,122}} = 180,6 \frac{\text{m}}{\text{min}} = 181 \frac{\text{m}}{\text{min}}$$

DRILLING

Consider a machining cycle to obtain a through hole with a diameter of 30 mm and a depth of 60 mm with a tolerance grade equal to IT8. The machining cycle involves a drilling operation with a twist drill bit with a diameter equal to 28 mm and a finishing operation by means of a single-point cutting tool (boring). The machining operation are carried out on a machine tool with a nominal power equal to 7,5 kW and an efficiency equal to 90%. Determine the cutting force, the cutting power, and the cutting time for both the machining operations. Are the machining operations feasible on the selected machine tool?

Drilling data:

- Cutting speed: $v_c = 290$ m/min
- Feed: $f = 0,10$ mm/revolution
- Cutting pressure: $k_c = 2100$ MPa
- Point angle: $\varepsilon = 160^\circ$

Finishing data:

- Cutting speed: $v_c = 325$ m/min
- Feed: $f = 0,05$ mm/revolution
- Single-point cutting tool: $Z = 1$
- Cutting pressure: $k_c = 2100$ MPa

Solution

Twist drilling:

$$A_D = \frac{f \cdot D}{4} = \frac{0,1 \cdot 28}{4} = 0,7 \text{ mm}^2$$

$$F_c = k_c A_D = 2100 \cdot 0,7 = 1470 \text{ N}$$

$$T_c = F_c \frac{D/2}{1000} = 1470 \frac{28/2}{1000} = 20,58 \text{ Nm}$$

$$n = \frac{1000 \cdot v_c}{\pi D} = \frac{1000 \cdot 290}{\pi \cdot 28} = 3297 \text{ RPM}$$

$$\omega = \frac{2\pi n}{60} = \frac{2\pi \cdot 3297}{60} = 345,3 \frac{\text{rad}}{\text{s}}$$

$$P_c = T_c \omega = 20,58 \cdot 345,3 = 7106 \text{ W} = 7,1 \text{ kW}$$

The power limit is $P_{c,limit} = E \cdot P_e = 0,9 \cdot 7,5 = 6,75 \text{ kW}$, therefore the drilling operation is not feasible.

In order to determine the cutting time, we need to compute the overall cutting length considering the conical shape of the drill bit end. Therefore:

$$A = \frac{D/2}{\tan \frac{\varepsilon}{2}} = \frac{28/2}{\tan \frac{160^\circ}{2}} = 2,47 \text{ mm}$$

$$L_c = H + A = 60 + 2,47 = 62,47 \text{ mm}$$

$$t_c = \frac{L_c}{v_f} = \frac{L_c}{f \cdot n} = \frac{62,47}{0,1 \cdot 3297} = 0,1895 \text{ min} = 11,37 \text{ s}$$

Finishing (boring):

This operation is a single-point cutting tool operation (similar to the internal cylindrical turning):

$$a_p = \frac{D_{final} - D_{drilling}}{2} = \frac{30 - 28}{2} = 1 \text{ mm}$$

$$A_D = f \cdot a_p = 0,05 \cdot 1 = 0,05 \text{ mm}^2$$

$$F_c = k_c A_D = 2.100 \cdot 0,05 = 105 \text{ N}$$

$$P_c = F_c v_c = 105 \cdot \frac{325}{60} = 568,75 \text{ W} = 0,57 \text{ kW}$$

The power limit is $P_{c,limit} = E \cdot P_e = 0,9 \cdot 7,5 = 6,75 \text{ kW}$, therefore the finishing operation is feasible.

$$n = \frac{1000 \cdot v_c}{\pi D} = \frac{1000 \cdot 325}{\pi \cdot 30} = 3.448 \text{ RPM}$$

$$t_c = \frac{L_c}{v_f} = \frac{H}{f \cdot n} = \frac{60}{0,05 \cdot 3.448} = 0,348 \text{ min} = 20,88 \text{ s}$$